

LECTURE 3

GROUP AND RING THEORY

Theorem 3.5 (First Sylow Theorem). *Let G be a finite group, and let p be a prime. If p^m divides $|G|$, then G has a subgroup of order p^m .*

Proof. If $m = 0$, the result is trivial, so we assume $m > 0$. We proceed by induction on $n = |G|$. If $n = 1$, the result is trivially true. So we assume that the result holds for all groups of order less than n .

If the order of the centre $Z(G)$ of G is divisible by p , then it follows from the Fundamental Theorem of Finite Abelian Groups that $Z(G)$ contains an element a of order p . The cyclic group $C = \langle a \rangle$ is clearly normal in G , and $|G/C| = n/p$, which is divisible by p^{m-1} . Therefore, by induction there exists $H/C \leq G/C$ with $|H/C| = p^{m-1}$, proving the theorem in this case since $|H| = p^m$.

Now consider the case when $p \nmid |Z(G)|$. Recall that the class equation of G is

$$n = |G| = |Z(G)| + \sum_a |G : C_G(a)|$$

where the summation runs over one element from each conjugacy class having more than one element. Now $p \mid n$, but $p \nmid |Z(G)|$. Hence $p \nmid |G : C_G(a)|$ for some $a \in G \setminus Z(G)$. This implies that $p^m \mid |C_G(a)|$ and $|C_G(a)| < |G|$. By the induction hypothesis $C_G(a)$ has a subgroup of order p^m , and this yields the required subgroup of G . $\square$

A special case of this theorem is:

Corollary 3.6 (Cauchy's Theorem). *If the order of a finite group G is divisible by a prime number p , then G has an element of order p .*

Proof. By the First Sylow Theorem there exists a subgroup of order p . Therefore there are at least $p - 1$ elements of order p (namely, all but the identity element of the cyclic subgroup of order p). $\square$

Corollary 3.7. *A finite group G is a p -group if and only if its order is a power of p .*

Proof. If G is of p -power order, then each element trivially has order a power of p . So let us assume that each element in G has order a power of p and prove that the order of G is a power of p . Suppose for a contradiction that $q \neq p$ is a prime number dividing the order of G . Then by Cauchy's Theorem there exists an element of order q , a contradiction. Hence G is of order a power of p . $\square$

Theorem 3.8 (Second Sylow Theorem). *Let G be a finite group, and let p be a prime. Then all Sylow p -subgroups of G are conjugate.*

Proof. Let K be a Sylow p -subgroup in G . Let $\text{ccl}(K) = \{xKx^{-1} \mid x \in G\}$ be the family of subgroups of G conjugate to K . Then by the Orbit-Stabiliser Theorem,

$$|\text{ccl}(K)| = |G : N_G(K)|.$$

Since K is a Sylow p -subgroup, we have $(|G : N_G(K)|, p) = 1$. Therefore

$$(3.1) \quad (|\text{ccl}(K)|, p) = 1.$$

Next let H be any Sylow p -subgroup. We want to show that H is conjugate to K . Regard the set $\text{ccl}(K)$ as a H -set by conjugation. Then for any $L \in \text{ccl}(K)$, let $\text{ccl}_H(L) = \{hLh^{-1} \mid h \in H\}$, the orbit of L . Obviously $\text{ccl}_H(L) \subseteq \text{ccl}(K)$, and

$$(3.2) \quad \text{ccl}(K) = \bigcup_{L \in \text{ccl}(K)} \text{ccl}_H(L)$$

is a disjoint union, where the union runs over one element L from each orbit (= conjugacy class $\text{ccl}_H(L)$).

From Example 1.29, we have

$$(3.3) \quad |\text{ccl}_H(L)| = [H : H \cap N_G(L)] = p^e, \quad e \geq 0,$$

because H is a Sylow p -subgroup. We first show that $p^e = 1$ if and only if $H = L$. If $H = L$, then trivially $p^e = 1$. Conversely,

$$p^e = 1 \implies H = H \cap N_G(L) \implies H \subseteq N_G(L) \implies HL < G \text{ and } L \leq HL.$$

On the other hand, by the Second Isomorphism Theorem,

$$HL/L \cong H/(H \cap L) \implies HL/L \text{ is a } p\text{-group} \implies HL = L,$$

since L is a Sylow p -subgroup; indeed, we observe that $p \nmid |G|/|L|$, and as $|HL|/|L| \mid |G|/|L|$, the fact that HL/L is a p -group implies that HL/L is the trivial group. So our assertion that $p^e = 1$ if and only if $H = L$ is established.

By (3.2) and (3.3) we have

$$(3.4) \quad |\text{ccl}(K)| = \sum p^e.$$

Then (3.1) implies that in $\sum p^e$ at least one term p^e must be equal to 1, and this in turn implies, by what we established in the last paragraph, that H must be equal to some conjugate L of K .

[Here is a shorter alternative proof: Denote by $\text{Syl}_p(G)$ the set of all Sylow p -subgroups of G . Let $S, H \in \text{Syl}_p(G)$. Consider the action of S by translation on the set of cosets G/H . From the Orbit-Stabiliser Theorem, we know that each S -orbit is a power of p , since S is a p -subgroup. For i a p -power, write λ_i for the number of S -orbits of cardinality i . Then

$$|G/H| = \lambda_1 + p\lambda_p + p^2\lambda_{p^2} + \cdots + p^m\lambda_{p^m},$$

where m is such that $p^m = |S|$. Since $(|G/H|, p) = 1$, it follows that $\lambda_1 > 0$. Thus there is a coset gH such that $s \cdot gH = gH$ for each $s \in S$. In other words, we have

$$sgh_1 = gh_2$$

for some $h_1, h_2 \in H$. Rearranging gives

$$s = gh_2h_1^{-1}g^{-1} \in gHg^{-1}.$$

As $s \in S$ was arbitrary, we obtain that $S \subseteq gHg^{-1}$. Since $|S| = |H|$, we have $S = gHg^{-1}$, as wanted. $\square$

Theorem 3.9 (Third Sylow Theorem). *Let G be a finite group, and let p be a prime. Then the number n_p of all Sylow p -subgroups of G satisfies $n_p \mid |G|$ and $n_p \equiv 1 \pmod{p}$.*

Proof. As all Sylow p -subgroups of G are conjugate to some fixed Sylow p -subgroup K (from the Second Sylow Theorem), it follows that their number n_p is equal to $|\text{ccl}(K)| = |G|/|N_G(K)|$. Hence $n_p \mid |G|$.

It is also clear that there is one and only one term in the sum $\sum p^e$ (in (3.4)) that is equal to 1, so we have from (3.4) that the number of distinct conjugates of K is

$$n_p = 1 + \sum_{e>0} p^e = 1 + kp \equiv 1 \pmod{p}.$$

[Here is an alternative proof, that is more in line with the alternative proof of the Second Sylow Theorem above: Recall that $\text{Syl}_p(G)$ is the set of all Sylow p -subgroups of G . Let G act on $\text{Syl}_p(G)$ by conjugation. Take $P \in \text{Syl}_p(G)$. The Second Sylow Theorem gives that the orbit of P has cardinality n_p , so by the Orbit-Stabiliser Theorem, we get $n_p = [G : G_P]$, so clearly $n_p \mid |G|$.

Next let P act on $\text{Syl}_p(G)$ by conjugation, and let F denote the set of fixed points of this action. Choose $Q \in F$ and note that $Q = x^{-1}Qx$ for each $x \in P$, and so $P \leq N_G(Q)$. Appealing again to the Second Sylow Theorem, we have in particular that P and Q are conjugate in $N_G(Q)$. As $Q \trianglelefteq N_G(Q)$, it follows that $P = Q$. Hence $F = \{P\}$. We now claim that $|F| \equiv |\text{Syl}_p(G)| \pmod{p}$, from which the result then follows.

So it remains to prove the claim. From the Orbit-Stabiliser Theorem, any point $x \in \text{Syl}_p(G) \setminus F$ will lie in an orbit of cardinality $|P|/|P_x|$, which is a multiple of p by assumption. The claim then follows, upon writing $|\text{Syl}_p(G)|$ as a sum of $|P \cdot x|$ taken over all the distinct orbits $P \cdot x$, and then reducing modulo p . $\square$

Corollary 3.10. *A Sylow p -subgroup of a finite group G is unique if and only if it is normal.*

3.2. Applications.

Example 3.11. If a group G of order p^n contains exactly one subgroup of order p^k for each $k \in \{1, \dots, n-1\}$, then G is cyclic.

Solution: Let $H \leq G$ be a subgroup of order p^{n-1} . By the First Sylow Theorem, the subgroup H contains subgroups of orders $p, p^2, \dots, p^{n-2}$. Since these subgroups are also subgroups of G , by the assumption of G it follows that all proper subgroups of G are contained in H . Now if $a \in G \setminus H$, then the order of a must be p^n , otherwise a will generate a subgroup K of order $< p^n$. This would mean $a \in H$, as $a \in K \subseteq H$. Thus $o(a) = p^n$, and G is therefore a cyclic group generated by a .

Example 3.12. If d is a divisor of the order n of a finite abelian group A , then A contains a subgroup of order d .

Solution: If $n = p_1^{e_1} \cdots p_r^{e_r}$ is a decomposition of n into a product of distinct prime powers, then $A = B(p_1) \oplus \cdots \oplus B(p_r)$ where

$$|B(p_i)| = p_i^{e_i}, \quad \text{for } 1 \leq i \leq r;$$

cf. the Fundamental Theorem of Finite Abelian Groups.

Now write $d = p_1^{f_1} \cdots p_r^{f_r}$ for appropriate $0 \leq f_i \leq e_i$. By the First Sylow Theorem, each $B(p_i)$ contains a subgroup $B'(p_i)$ of order $p_i^{f_i}$. Then $S = B'(p_1) \oplus \cdots \oplus B'(p_r)$ is a subgroup of order d , as required.

Note that the above example proves the converse of Lagrange's Theorem for abelian groups.

Example 3.13. Prove that there are no simple groups of orders 63, 56, and 36.

Solution: We have $63 = 3^2 \cdot 7$. By the Third Sylow Theorem, the number of Sylow 7-subgroups is $n_7 = 1 + 7k$, with $1 + 7k$ dividing $3^2 \cdot 7$. It then follows that $1 + 7k$ divides 3^2 . This yields $k = 0$. Therefore, there is a unique Sylow 7-subgroup that must be a normal subgroup. Hence no group of order 63 can be simple.

Similarly, for groups of order 56, the number of Sylow 7-subgroups $n_7 = 1 + 7k$ must divide $2^3 \cdot 7$. Thus $(1 + 7k) \mid 2^3$. This yields $n_7 = 1$ or 8. If $n_7 = 1$, we are done as before. So we suppose that $n_7 = 8$. Then there are $8 \cdot 6 = 48$ elements of order 7. Then the remaining eight elements must form a unique Sylow 2-subgroup. So again we have a normal subgroup, and therefore G cannot be a simple group.

The non-simplicity of groups of order 36 may be proved similarly. Here is an alternative method, which is sometimes easier. Let H be a Sylow 3-subgroup of G . By Theorem 1.12, there exists a homomorphism $\varphi : G \rightarrow S_4$, as $|G : H| = 4$. Note $\ker \varphi = \{e\}$ implies $G \hookrightarrow S_4$, which is not possible since $|G| = 36$ and $|S_4| = 24$. Also $\ker \varphi = \cap_{x \in G} xHx^{-1} \neq G$. Thus $\ker \varphi$ is a non-trivial normal and proper subgroup of G and so G is not simple.

Example 3.14. Every finite p -group is isomorphic to some subgroup of the group of upper unitriangular matrices over $\mathbb{F}_p$.

Solution: Note that

$$\mathrm{GL}_n(\mathbb{F}_p) = \prod_{k=0}^{n-1} (p^n - p^k) = p^{\frac{n(n-1)}{2}} \prod_{k=0}^{n-1} (p^{n-k} - 1).$$

Since $p \nmid (p^{n-k} - 1)$ for all $k \in \{0, 1, \dots, n-1\}$, a Sylow p -subgroup of $\mathrm{GL}_n(\mathbb{F}_p)$ has order $p^{\frac{n(n-1)}{2}}$.

The upper unitriangular group $U_n(\mathbb{F}_p)$ is a subgroup of $\mathrm{GL}_n(\mathbb{F}_p)$; indeed, it is closed under multiplication and all its matrices have determinant 1. Note that an upper unitriangular matrix is of the form

$$\begin{pmatrix} 1 & * & \cdots & \cdots & * \\ 0 & 1 & * & \cdots & * \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & * \\ 0 & \cdots & \cdots & 0 & 1 \end{pmatrix}$$

where $*$ stands for an arbitrary element of $\mathbb{F}_p$. Since there are no restrictions on $*$, all possibilities occur, yielding $p^{\frac{n(n-1)}{2}}$ distinct upper unitriangular matrices. Hence $U_n(\mathbb{F}_p)$ is a Sylow p -subgroup of $\mathrm{GL}_n(\mathbb{F}_p)$.

If $|G| = p^k$ for some $k \geq 0$, writing $n := p^k$ we can embed G into S_n , which in turn can be embedded into $\mathrm{GL}_n(\mathbb{F}_p)$ via the permutation matrices. Thus G is a p -subgroup of $\mathrm{GL}_n(\mathbb{F}_p)$, and hence it is contained in some Sylow p -subgroup H of $\mathrm{GL}_n(\mathbb{F}_p)$. By the Second Sylow Theorem, the subgroup H is conjugate to $U_n(\mathbb{F}_p)$, and the result follows.

3.3. Further applications to groups of orders p^2 , pq (non-examinable for the oral exam). Recall that $\langle x \rangle$ is the cyclic group generated by x , and more generally $\langle x, y \rangle$ is the group generated by x and y . Throughout this section p and q are distinct primes.

3.3.1. Groups of order p^2 . We have already shown (Exercise Sheet 2) that a group of order p^2 is abelian. Hence, by the Fundamental Theorem of Finite Abelian Groups, there are exactly two groups (up to isomorphism) of order p^2 :

- (i) the abelian group of type (p^2) , and
- (ii) the abelian group of type (p, p) .

[Recall that an abelian group of type $(n_1, n_2, \dots, n_k)$ is isomorphic to $C_{n_1} \times C_{n_2} \times \dots \times C_{n_k}$.]

3.3.2. Groups of order pq , for $q > p$. Let G be a group of order pq . Then the number of Sylow q -subgroups of G is of the form $1 + \lambda q$, which divides pq . This gives $1 + \lambda q = 1$, and thus there is exactly one Sylow q -subgroup, which, being of order q , is cyclic. Let $\langle b \rangle$, where $b^q = 1$, be this unique Sylow q -subgroup.

Also the number of Sylow p -subgroups is of the form $1 + \mu p$, which divides pq . This gives $1 + \mu p \in \{1, q\}$.

Case (i). Suppose the number of Sylow p -subgroups in G is 1, and let this be $\langle a \rangle$, where $a^p = 1$. Since $\langle a \rangle$ and $\langle b \rangle$ are normal in G , and $\langle a \rangle \cap \langle b \rangle = \{e\}$, we have $G = \langle a \rangle \times \langle b \rangle$. Since $o(ab) = pq$, we have $G = \langle ab \rangle$, and thus G is cyclic.

Case (ii). The number of Sylow p -subgroups is q . In this case $q \equiv 1 \pmod{p}$. Let $\langle a \rangle$ be one of the Sylow p -subgroups of G , so $a^p = 1$. Then $\langle a, b \rangle$ contains $\langle a \rangle$ of order p and $\langle b \rangle$ of order q . Thus $|\langle a, b \rangle| = pq$, whence $G = \langle a, b \rangle$. As $\langle b \rangle \trianglelefteq G$, we have $a^{-1}ba = b^r$ for some integer r . Then $r \not\equiv 1 \pmod{q}$, for otherwise $ba = ab$, and G is abelian, which in turn implies that the number of Sylow p -subgroups is 1, a contradiction. Thus G is generated by a and b with defining relations

$$(3.5) \quad a^p = 1 = b^q, \quad a^{-1}ba = b^r.$$

Observe that the integer r in (3.5) is a solution of the congruence equation

$$(3.6) \quad z^p \equiv 1 \pmod{q}.$$

Indeed,

$$\begin{aligned} a^{-1}ba = b^r &\implies a^{-1}b^ra = (a^{-1}ba)^r = (b^r)^r = b^{r^2} \\ &\implies a^{-1}(a^{-1}ba)a = b^{r^2} \implies a^{-2}ba^2 = b^{r^2} \\ &\implies a^{-p}ba^p = b^{r^p} \quad \text{by induction} \\ &\implies b = b^{r^p} \implies r^p \equiv 1 \pmod{q}. \end{aligned}$$

Conversely, if r is a solution of the congruence equation (3.6), then the defining relations (3.5) determine a group consisting of the pq distinct elements a^ib^j , for $0 \leq i < p-1$ and $0 \leq j < q-1$.

We note that the solutions of $z^p \equiv 1 \pmod{q}$ with $z \not\equiv 1 \pmod{q}$ are $r, r^2, \dots, r^{p-1}$, and all yield the same group, because replacing a by a^i as a generator of $\langle a \rangle$ replaces r by r^i . Note also that by Fermat's Little Theorem, we have p divides $q-1$.

We conclude that there are at most two groups of order pq :

- (i) the cyclic group of order pq , and
- (ii) the non-abelian group generated by a and b with defining relations $a^p = 1 = b^q$, $a^{-1}ba = b^r$, $r^p \equiv 1 \pmod{q}$, $r \not\equiv 1 \pmod{q}$, provided p divides $q-1$.